

REST API using Spring Boot

What is REST API?

REST API is used to communicate between client and server. It exchanges data using **JSON**.

HTTP Methods

- **GET** → Fetch data
- **POST** → Create data
- **PUT** → Update data
- **DELETE** → Remove data

What is Endpoint?

Endpoint is a URL used to access API.

Example: /users

Spring Boot Annotations

- @RestController
- @GetMapping
- @PostMapping

Simple Example (Mastercode Notes)

@RestController

@GetMapping("/hello")

```
public String hello() {
    return "Hello World";
}
```

Working Flow

Client → API Request → Controller → Response (**JSON**)

JSON Format

```
{
  "name": "Sagar"
}
```

Key Idea

REST API = Data Communication

Use in Industry

Used in web apps, mobile apps, and backend systems.



Spring Boot + JPA (Database Integration)

What is JPA?

JPA (Java Persistence API) is used to interact with **database** in Java applications.

What is ORM?

ORM (Object Relational Mapping) maps Java objects to database tables.

Entity Class

Entity represents a table in the **database**.

Annotation: **@Entity**

Repository Layer

Repository is used to perform **database** operations.

Example: **JpaRepository**

Database Configuration

Configured in application.properties

Example: spring.datasource.url

Simple Flow

Controller → Service → **Repository** → **Database**

Basic Example (Mastercode Notes)

@Entity

```
class User {
    int id;
    String name;
}
```

Key Idea

JPA = Database Handling Made Easy

Use in Projects

Used to store, update, and fetch data from **database**.



Spring Security (Authentication & Authorization)

What is Spring Security?

Spring Security is used to secure applications and APIs. It handles login and access control.

Authentication

Authentication means verifying user identity.

Example: Username + Password login

Authorization

Authorization means giving permission to users.

Example: Admin vs User roles

Key Features

- Login system
- Password encryption
- Role-based access
- Secure APIs

Basic Flow

User → Login → Authenticate → Authorize → Access

Spring Security Components

- UserDetails
- Password Encoder
- Security Config

Simple Example (Concept)

User enters credentials → System verifies → Access granted or denied

Key Idea

Authentication = Who you are

Authorization = What you can do

Use in Real Projects

Used in banking apps, web apps, and secure APIs.



Microservices Architecture (Spring Boot)

What is Microservices?

Microservices is an architecture where application is divided into small independent services.

Features of Microservices

- Independent services
- Scalable
- Easy to maintain
- Faster development

Monolithic vs Microservices

Monolithic → Single large application

Microservices → Multiple small services

How Microservices Work

Each service runs independently and communicates using APIs.

Communication Between Services

- REST API
- HTTP requests

Simple Flow

Client → API Gateway → Services → Database

Advantages

- Better scalability
- Fault isolation
- Easy deployment

Key Idea

Microservices = Divide & Manage

Use in Industry

Used in large systems like Netflix, Amazon, etc.



Spring Boot Workflow (End-to-End)

What is Workflow?

Workflow is the step-by-step process of how a request is handled in a Spring Boot application.

Steps in Workflow

1. Client sends request
2. Controller receives request
3. Service processes logic
4. Repository interacts with database
5. Response is returned

Controller Layer

Handles incoming requests and sends response.

Service Layer

Contains business logic and processing.

Repository Layer

Communicates with database to fetch or store data.

Database

Stores application data.

Simple Flow Diagram

Client → Controller → Service → Repository → Database → Response

Key Idea

Layered Architecture = Clean Code

Use in Projects

Used in all backend systems to handle requests efficiently.



Real-World Use Cases of Spring Boot

What is Use Case? -----

Use case means real-world application of a technology.

Web Applications -----

Used to build dynamic **web applications**.

Example: E-commerce websites

REST APIs -----

Used to create **REST APIs** for communication between client and server.

Microservices Systems -----

Used in distributed systems with multiple services.

Enterprise Applications -----

Used in large-scale business applications.

Cloud Applications -----

Used in cloud-based systems for scalability.

Banking & Finance -----

Used in secure systems for transactions and data.

Simple Flow -----

Client → **Backend** → Database → Response

Key Idea

Spring Boot = Real Industry Tool

Use in Industry

Widely used in companies for **backend** development.

